

ASCII MASTER FLOW – PANEL 1/12: Satellite service taxonomy

EARTH OBSERVATION -> sensed Earth data and derived products

COMMUNICATION -> relay through transponders and ground networks

METEOROLOGY -> imaging, sounding, relay and warning support

NAVIGATION -> position, navigation and timing signals

RULE -> payload and service define the class

ASCII MASTER FLOW – PANEL 2/12: Orbit-service matrix

GEO -> continuous regional view and fixed ground geometry

IGSO / GSO -> daily-period geometry with changing sky position

POLAR LEO -> repeated high-inclination Earth coverage

SUN-SYNCHRONOUS LEO -> near-constant local-solar-time imaging

RULE -> orbit enables service but does not name the payload

ASCII MASTER FLOW – PANEL 3/12: End-to-end segment chain

SPACE SEGMENT -> satellite bus and mission payload

CONTROL SEGMENT -> tracking, command and health management

PROCESSING SEGMENT -> correction, calibration or data products

USER SEGMENT -> receiver, department or operational workflow

OUTCOME -> useful service only after all segments function

ASCII MASTER FLOW – PANEL 4/12: Observation-to-decision rail

SENSOR -> records reflected or emitted energy

DOWNLINK -> sends measurements to a ground station

PROCESSING -> calibration, correction and product generation

USER AGENCY -> interprets the product

DECISION -> mapping, crop, water, warning or response action

ASCII MASTER FLOW – PANEL 5/12: Communication relay loop

UPLINK -> terrestrial station sends a signal

TRANSPONDER -> receives, shifts, amplifies and retransmits

DOWNLINK -> signal reaches a service footprint

GROUND NETWORK -> routes content to final users

RULE -> satellite capacity is not the same as delivered access

ASCII MASTER FLOW – PANEL 6/12: NavIC system map

REGIONAL CONSTELLATION -> independent Indian PNT source

GEO PLUS IGSO DESIGN -> nominal regional geometry

CONTROL SEGMENT -> orbit and clock monitoring

USER RECEIVER -> measures signal travel time

RULE -> nominal design is not dated functional health

ASCII MASTER FLOW – PANEL 7/12: NavIC service and signal stack

SPS -> open positioning service

RS -> encrypted service for authorised users

L5 / S -> original signal layers in owner

L1 -> mass-market interoperability direction

RULE -> payload support does not prove device adoption

ASCII MASTER FLOW – PANEL 8/12: Constellation-health firewall

LAUNCHED -> cumulative mission count

INTENDED ORBIT -> orbit-raising completed

OPERATIONAL -> spacecraft performing a stated function

PNT CAPABLE -> full navigation service contribution

MESSAGE BROADCAST -> useful but not equivalent to PNT

ASCII MASTER FLOW – PANEL 9/12: GAGAN correction chain

REFERENCE STATIONS -> measure GNSS error

MASTER CONTROL -> computes correction and integrity

UPLINK STATIONS -> send augmentation message

GEO PAYLOAD -> rebroadcasts over a wide area

AIRCRAFT RECEIVER -> applies correction and checks integrity

ASCII MASTER FLOW – PANEL 10/12: Accuracy-integrity gate

ACCURACY -> estimated position close to true position

INTEGRITY -> warning when information must not be used

TIME TO ALERT -> warning must arrive within a safe interval

AVIATION USE -> certification governs operational acceptance

RULE -> a precise signal without integrity is unsafe

MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

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ASCII MASTER FLOW – PANEL 11/12: SBAS-GBAS institution map

SBAS -> wide-area corrections rebroadcast from satellites

GBAS -> local airport corrections broadcast from the ground

ISRO + AAI -> GAGAN development partnership

AAI -> aviation operational role

DGCA -> aviation certification role

CLOSE DISTINCTION: NavIC is India's regional satellite-navigation system, while GAGAN is...

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ASCII MASTER FLOW – PANEL 12/12: Public-value answer spine

CLASSIFY -> observation communication meteorology PNT or augmentation

TRACE -> space ground processing and user segments

DISTINGUISH -> NavIC constellation from GAGAN SBAS

TEST -> standards adoption integrity privacy jamming and spoofing

VERIFY -> dated health frequency coverage accuracy and certification

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State constellation/service area, orbit,...

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